Author: To begin with, let us note that if the sequence (y_n) has a limit a, the sequence $(-y_n)$ is also convergent and its limit is $-a$.
This follows from an easily proved equality $\boxed{\lim_{n\to\infty}(cy_n) = c\lim_{n\to\infty} y_n}$ where c is a constant.

- Assume now that a sequence (y_n+z_n) is convergent to A, and that (y_n) is also convergent and its limit is a. Let us apply the theorem on the sum of convergent sequences to the sequences (y_n+z_n) and $(-y_n)$. As a result, we obtain that the sequence $(y_n+z_n-y_n)$, i.e. (z_n) is also convergent, with the limit $A-a$.

Scan: **Claim:** $\lim_{n\to\infty}(cy_n) = c\lim_{n\to\infty} y_n$

Proof: Let $\lim_{n\to\infty}(cy_n) = a$. By defn. of limit, $\forall \varepsilon > 0$, $\varepsilon \in \mathbb{R}$, $\exists N_1 \in \mathbb{N}$ s.t.

$\forall n \in \mathbb{N}, n > N_1$, $|cy_n - a| < \varepsilon \Rightarrow \boxed{a-\varepsilon < cy_n < a+\varepsilon}$ (Eqn 1)

Case 1: $c \neq 0$. $\therefore \frac{a-\varepsilon}{c} < y_n < \frac{a+\varepsilon}{c} \Rightarrow \frac{a}{c} - \frac{\varepsilon}{c} < y_n < \frac{a}{c} + \frac{\varepsilon}{c}$

$\Rightarrow \boxed{\frac{a}{c} - \varepsilon' < y_n < \frac{a}{c} + \varepsilon'}$, where $\varepsilon' = \frac{\varepsilon}{c}$.

$\Rightarrow |y_n - \frac{a}{c}| < \varepsilon'$. This holds true for any arbitrary ε'. !!

$\therefore \lim_{n\to\infty} y_n = \frac{a}{c}$ $\therefore c\lim_{n\to\infty} y_n = c\cdot\frac{a}{c} = a = \lim_{n\to\infty}(cy_n)$

Case 2: $c = 0$. $\therefore \forall y_n, cy_n = 0$ $\therefore \lim_{n\to\infty}(cy_n) = 0$

Also, $c\lim_{n\to\infty}(y_n) = 0\cdot\lim_{n\to\infty}(y_n) = 0$ $\square$

Reader: Indeed (z_n) cannot be divergent in this case.

Author: Very well. Let us discuss now one important particular case of convergent sequences, namely, the so called $\boxed{\text{infinitesimal sequence}}$, or simply, $\boxed{\text{infinitesimal}}$. This is the name which is given to a convergent sequence with a limit equal to zero. Sequences (7) and (9) from Dialogue One are examples of infinitesimals.

Note that to any convergent sequence (y_n) with a limit $\boxed{a}$ there corresponds an $\boxed{\text{infinitesimal sequence}}$ (α_n), where $\boxed{\alpha_n = y_n - a}$. That is why mathematical analysis is also called calculus of infinitesimals.